MUHAMMED BASIL CP

+91 9567738352 basilcp090@gmail.com LinkedIn Github

Education

Lovely Professional University, Punjab **2021 - 2025**

B.Tech in Computer Science & Engineering

TSS Vadakkangara , Kerala **2018 - 2020**

Higher Secondary Percentage: 88

Experience

Data Science Intern – CODO AI Innovations **July 2025 – Aug 2026**

- Performed data preprocessing, cleaning, and exploratory data analysis using Python, Pandas, and NumPy.
- Developed and evaluated classification and prediction models using Scikit-learn, Random Forest, LightGBM, and XGBoost.
- Worked on practical AI/NLP and data science problems, gaining experience in model development and deployment.
- Developed interactive data science applications using Streamlit.

Intern – Mental Health Companion Project **Jan'25 - Apr'25**

- Successfully completed a 3-month internship focused on developing the *Mental Health Companion* project..
- Collaborated on Mental Health Companion project by optimizing backend logic, fixing bugs, and improving security in API endpoints.
- Collaborated on resolving technical issues and improving the overall quality of the application.

Projects

Vakyam AI – Malayalam Learning Chatbot | Python, Rasa, Hugging Face | *Python, Rasa, Hugging Face* **Feb'25 – March'25**

- Built an interactive chatbot using reinforcement learning and rule-based grammar to teach Malayalam to beginners.
- Achieved 95.2% F1-score and 99.6% accuracy in improved English ↔ Malayalam translation with contextual responses and speech support.
- Deployed MPT-7B-Chat via Hugging Face for multilingual generation and used Rasa for efficient intent classification and dialogue management.

Malicious or Benign URL detection using ML: A ML PROJECT | *jupyter and streamlit* **Apr'24 – May'24**

- Developed a multi-class classification approach to distinguish between Benign, Phishing, Malware, and Defacement URLs.
- Engineered and extracted numerical features from raw URLs for optimized model input.
- Implemented an ensemble of Random Forest, LightGBM, and XGBoost, improving classification accuracy to 95%+ and enabling real-time detection via a Streamlit dashboard..

SecureScan – PII Redaction System for Documents | *python, DeBERTaV3, FastAPI, Next.js* **Jun'23 – July'23**

- Developed an AI-powered PII detection and redaction system using fine-tuned DeBERTaV3, Tesseract OCR, PyMuPDF and FastAPI to automatically identify and redact Aadhaar, PAN and passport numbers from documents and images.
- Integrated OCR (Tesseract), PDF processing (PyMuPDF), and a FastAPI + Next.js stack to automate redaction of sensitive PII in documents and images.
- Delivered high-precision PII detection in real-world government records, ensuring data privacy compliance and scalable deployment..

Technical Skills

Languages: Python, JAVA, JavaScript, HTML/CSS

Data & Visualization: MySQL, Power BI, Tableau, Pandas, NumPy

ML/Data: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM

Frameworks & Libraries: FastAPI, Rasa, Hugging Face Transformers, Streamlit, Node.js

Game Development: Unity, Unreal engine, Blender

Other Tools: Git/GitHub

Achievements

- Authored and presented the research paper "VAKYAM AI: A Reinforcement-Driven Malayalam Mastery" at AAIIC-2025 (Asian AI Innovations Conference), July 2025.
- 4 star in hackerrank
- Participated in National level camp for NSS

Certifications

Computer Architecture and Computer Organization Masterclass | *Udemy* | *May'23*

The world of computed Networking | *Udemy* | *Nov'23*

Introduction to Hardware and Operation system | *coursera* | *Junl'23*